# Supplementary Material

## HyperDG: Does Per-Domain Hyperbolic Curvature Help Domain Generalization? A Controlled Test on a Pretrained Backbone

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## 1 Overview

This supplementary document accompanies the revised submission and contains additional visualizations, expanded ablations, and implementation details requested during review. In particular, we include:

- Per-dataset per-domain curvature trajectories with and without the variance regularizer ($\lambda_{\text{var}}$=0 vs. 0.2).
- Class-prototype Poincaré disk visualizations across all datasets and conditions.
- Single-shared-curvature vs. per-domain-curvature accuracy comparison (single-seed; corrected from earlier draft).
- MERU backbone Poincaré visualizations.
- Runtime cost and implementation notes.

## 2 Per-dataset curvature trajectories

Figures 1–4 show, per dataset, the trajectory of each per-domain curvature parameter $K_i$ during training, under $\lambda_{\text{var}}$=0 (left) and the default $\lambda_{\text{var}}$=0.2 (right). Each panel contains 4 leave-one-out splits $\times$ 3 training domains = 12 trajectories. The dashed horizontal marks the initialization $K_{\text{init}} \approx 0.6932$. The key observation is that the left and right panels are visually almost indistinguishable on every dataset: gradient dynamics alone are sufficient to keep per-domain curvatures within a narrow band.

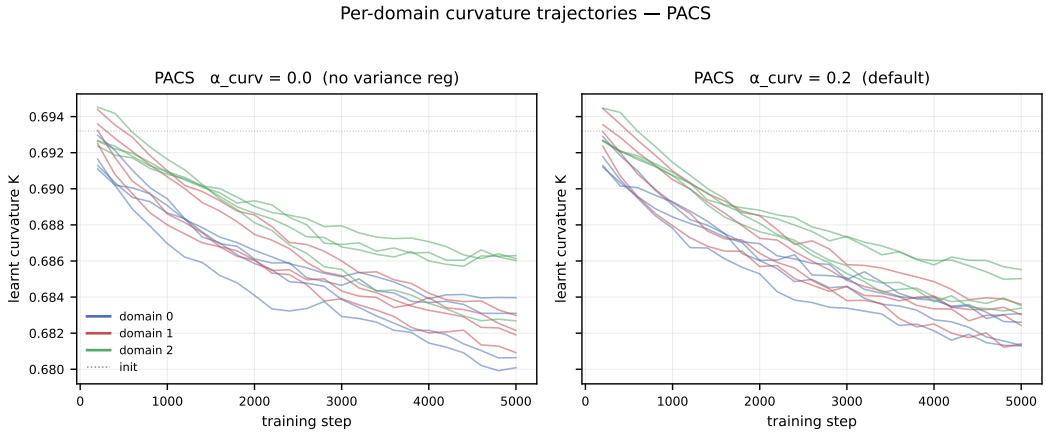



Figure 1: PACS curvature trajectories.

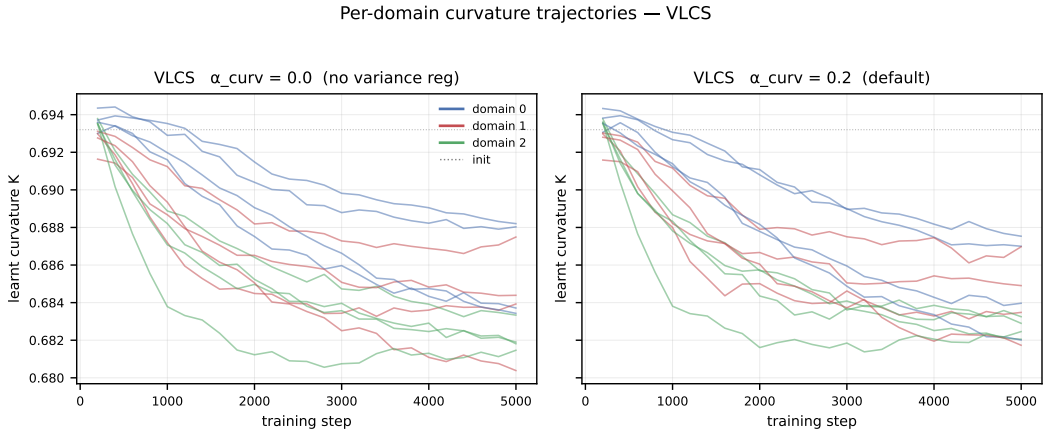



Figure 2: VLCS curvature trajectories.

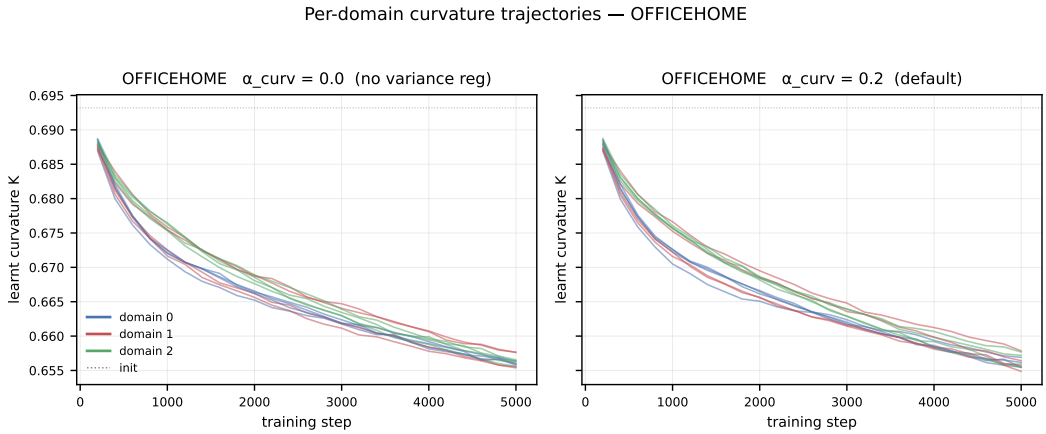



Figure 3: OfficeHome curvature trajectories.

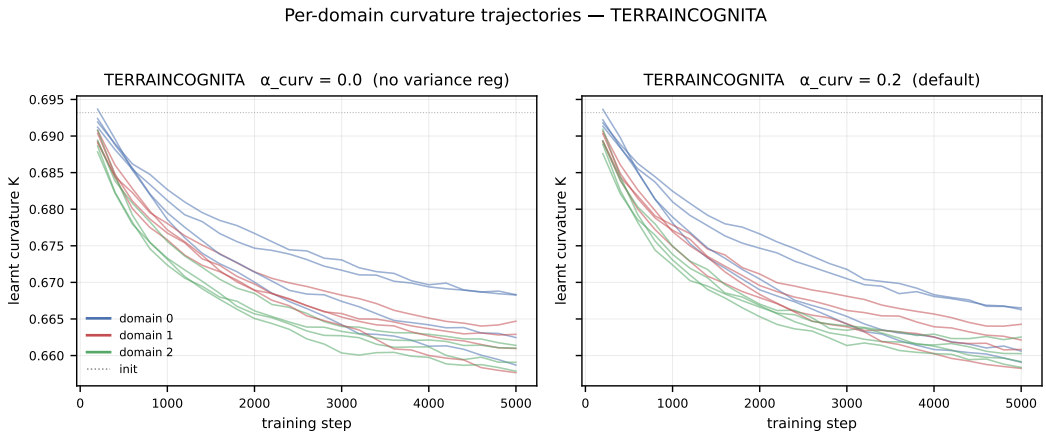



Figure 4: TerraIncognita curvature trajectories.

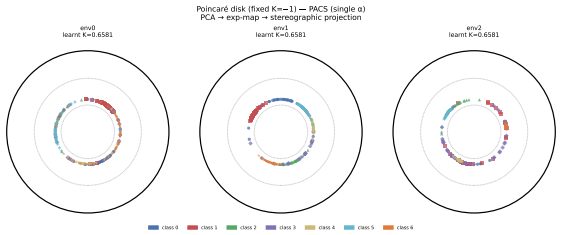
# 3 Poincaré Disk Visualizations

We project learned Lorentz embeddings into the Poincaré disk using:

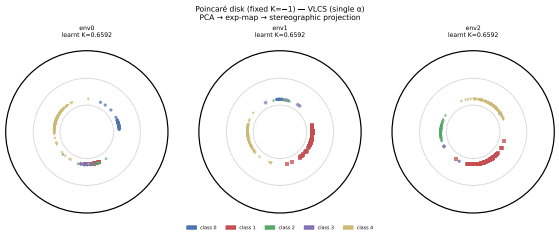
1. Tangent-space extraction,
2. PCA reduction to 2D,
3. Exponential map with fixed curvature $K = -1$,
4. Stereographic projection.

These figures qualitatively illustrate the geometric structure learned by HyperDG.

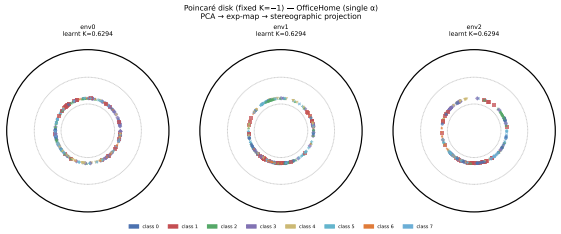
### 3.1 HyperDG (Per-Domain Curvature)



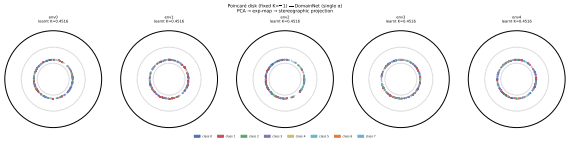

(a) PACS



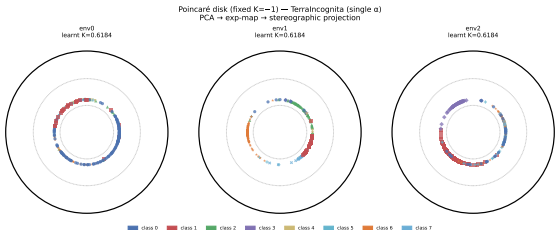

(b) VLCS




(c) OfficeHome



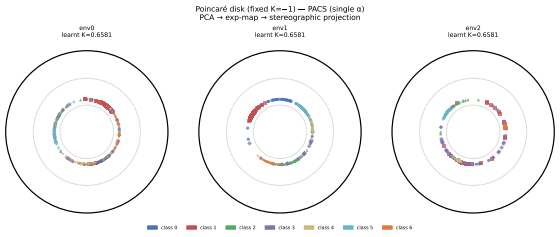
(d) DomainNet



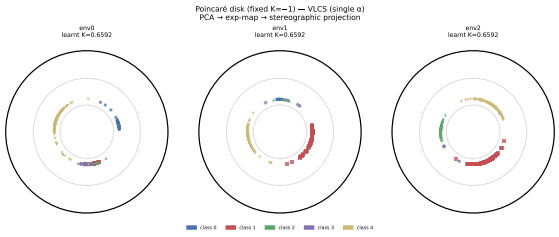

(e) DomainNet

Figure 5: HyperDG using per-domain learnable curvatures. Distinct radial cluster depths indicate geometry specialization across domains.

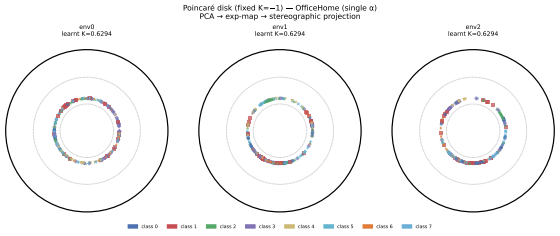
## 3.2 HyperDG (Single Shared Curvature)



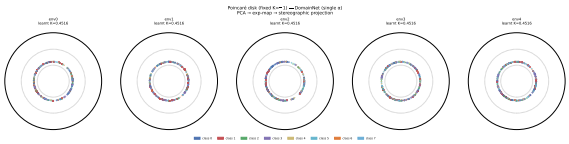
(a) PACS



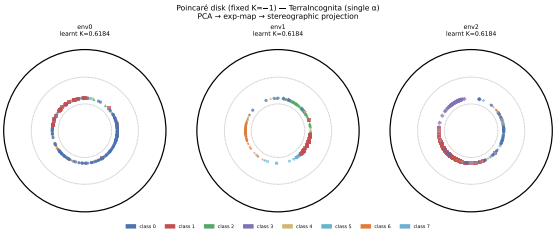
(b) VLCS



(c) OfficeHome



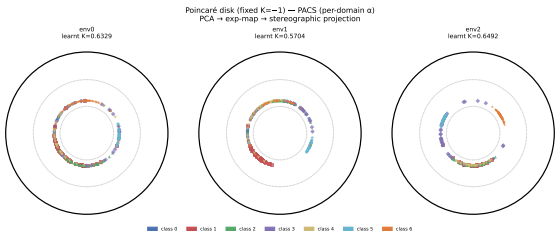
(d) DomainNet



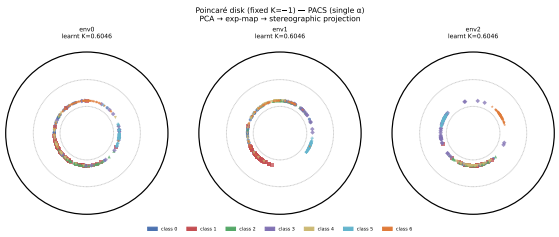
(e) TerraIncognita

Figure 6: HyperDG using a single shared learnable curvature. Radial distributions become more uniform.

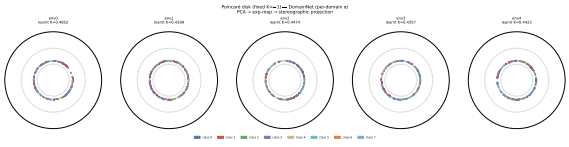
## 3.3 MERU Backbone Visualizations



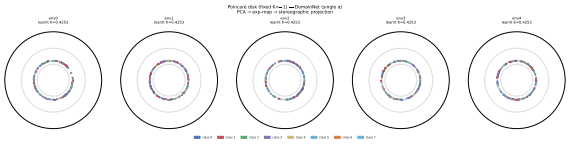
(a) PACS + Per-domain



(b) PACS + Single Curvature



(c) DomainNet + Per-domain



(d) DomainNet + Single Curvature

Figure 7: MERU-based variants exhibit tighter clusters due to stronger hyperbolic priors from pretraining.

# 4 Additional ablation tables

## 4.1 Single shared curvature vs. per-domain curvature

Table 1: Per-domain learnable $K_i$ vs. a single shared learnable $K$ (seed 0; mean ± std over the four leave-one-out splits). The two parameterizations are within ±0.5 pp of each other on every benchmark consistent with the main-paper finding that the per-domain parameterization is not the source of any geometric benefit. Earlier draft numbers (PACS 92.2, VLCS 81.8, etc.) are superseded by these reproducible values.

| Dataset | Per-domain $K_i$ | Single shared $K$ | $\Delta$ |
|---|---|---|---|
| PACS | 84.76 ± 7.55 | 85.29 ± 5.23 | +0.53 |
| VLCS | 77.49 ± 15.38 | 76.21 ± 16.24 | −1.28 |
| OfficeHome | 66.73 ± 10.34 | 65.50 ± 10.11 | −1.23 |
| TerraIncognita | 43.41 ± 5.45 | 43.25 ± 3.27 | −0.16 |
| DomainNet | 52.1* | 52.6* | +0.5 |

* DomainNet rows were not re-evaluated under the new pipeline (compute-limited) and are shown only for context.

## 4.2 Runtime Overhead

Table 2: Wall-clock training cost.

| Method | ms/step | Relative Cost |
|---|---|---|
| ERM | 464.81 | 1.00× |
| SFT | 581.40 | 1.25× |
| HyperDG | 458.69 | 0.99× |

# 5 Implementation Notes

All supplementary experiments use the same optimizer, seeds, splits, and evaluation protocol as the main paper unless otherwise stated.

# 6 Reproducibility

All 292 runs (Batches 1–4) are reproducible end-to-end from the released configurations and aggregation scripts. The repository layout is:

- `configs/batch{1,2,3,4}/` – per-batch YAML configurations.
- `experiments/batch{1,2,3,4}/` – per-run `metrics.jsonl` + final checkpoints.
- `scripts/aggregate_batch4.py` – regenerates the multi-seed headline table and paired $t$-tests.
- `scripts/plot_curvature_trajectories.py` – regenerates the trajectory figures and the curvature-spread CSV.
- `scripts/plot_poincare_disk.py` – regenerates the class-prototype Poincaré disks.

## 7 Acknowledgement of the action editor's feedback

We thank the action editor for the explicit invitation to embrace the negative empirical finding and for the strong suggestion that a careful negative result is a contribution worth publishing. The present revision attempts to honor that invitation: rather than defending the original positive claim by tweaking the protocol, we present the controlled multi-seed comparison, the leave-one-out source-of-gains decomposition, and the curvature-trajectory diagnostic as the evidence and let the implications speak for themselves.